Overview

We want to study the link between substance abuse and the suicide rate in the United States. For this we use data from 1999-2018. With the aim of carrying out a regression model to be able to predict the rates of future suicides dependent on drug addiction.

```
DESCRIPTIVES VARIABLES=Substance_Use Suicides
  /STATISTICS=MEAN STDDEV.
```

Descriptives

Descriptive Statistics

	N	Mean	Std. Deviation
Substance_Use	20	7.5550	.50598
Suicides	20	.00011920	.000011901
Valid N (listwise)	20		

```
CORRELATIONS
  /VARIABLES=Year Substance_Use Suicides
  /PRINT=TWOTAIL NOSIG
  /MISSING=PAIRWISE.
```

Correlations

Correlations

		Year	Substance_Use	Suicides
Year	Pearson Correlation	1	,976**	,975**
	Sig. (2-tailed)		,000	,000
	N	20	20	20
Substance_Use	Pearson Correlation	,976**	1	,955**
	Sig. (2-tailed)	,000		,000
	N	20	20	20
Suicides	Pearson Correlation	,975**	,955**	1
	Sig. (2-tailed)	,000	,000	
	N	20	20	20

**. Correlation is significant at the 0.01 level (2-tailed).

Interpretation

It can be noted that the correlations between the different variables are:
cor(Year/Substance_Use)=0.976.
cor(Year/Suicides)=0.975 .
cor(Suicided/Substance_Use)=0.955 .
We can thus conclude that the variables are quite related to each other with a significance of 0.01.

```
REGRESSION
  /MISSING LISTWISE
  /STATISTICS COEFF OUTS R ANOVA COLLIN TOL
  /CRITERIA=PIN(.05) POUT(.10)
  /NOORIGIN
  /DEPENDENT Suicides
  /METHOD=ENTER Year Substance_Use
  /PARTIALPLOT ALL
  /SCATTERPLOT=(*ZRESID ,*ZPRED)
  /RESIDUALS DURBIN HISTOGRAM(ZRESID) NORMPROB(ZRESID).
```

Regression

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	Substance_Use, Year[b]	.	Enter

a. Dependent Variable: Suicides
b. All requested variables entered.

Model Summary[b]

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Durbin-Watson
1	,975[a]	,950	,945	.000002800	,476

a. Predictors: (Constant), Substance_Use, Year
b. Dependent Variable: Suicides

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	,000	2	,000	163,142	,000[b]
	Residual	,000	17	,000		
	Total	,000	19			

a. Dependent Variable: Suicides

b. Predictors: (Constant), Substance_Use, Year

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.		
		B	Std. Error	Beta				
1	(Constant)	-,004	,001		-3,667	,002		
	Year	1,796E-6	,000	,893	3,615	,002		
	Substance_Use	1,979E-6	,000	,084	,341	,737		

Collinearity Diagnostics[a]

Model	Dimension	Eigenvalue	Condition Index	Variance Proportions		
				(Constant)	Year	Substance_Use
1	1	2,997	1,000	,00	,00	,00
	2	,003	33,226	,00	,00	,05
	3	2,053E-7	3821,176	1,00	1,00	,95

a. Dependent Variable: Suicides

Residuals Statistics[a]

	Minimum	Maximum	Mean	Std. Deviation	N
Predicted Value	.00010069	.00013760	.00011920	.000011603	20
Residual	-.000004087	.000004405	.000000000	.000002648	20
Std. Predicted Value	-1,596	1,585	,000	1,000	20
Std. Residual	-1,460	1,573	,000	,946	20

a. Dependent Variable: Suicides

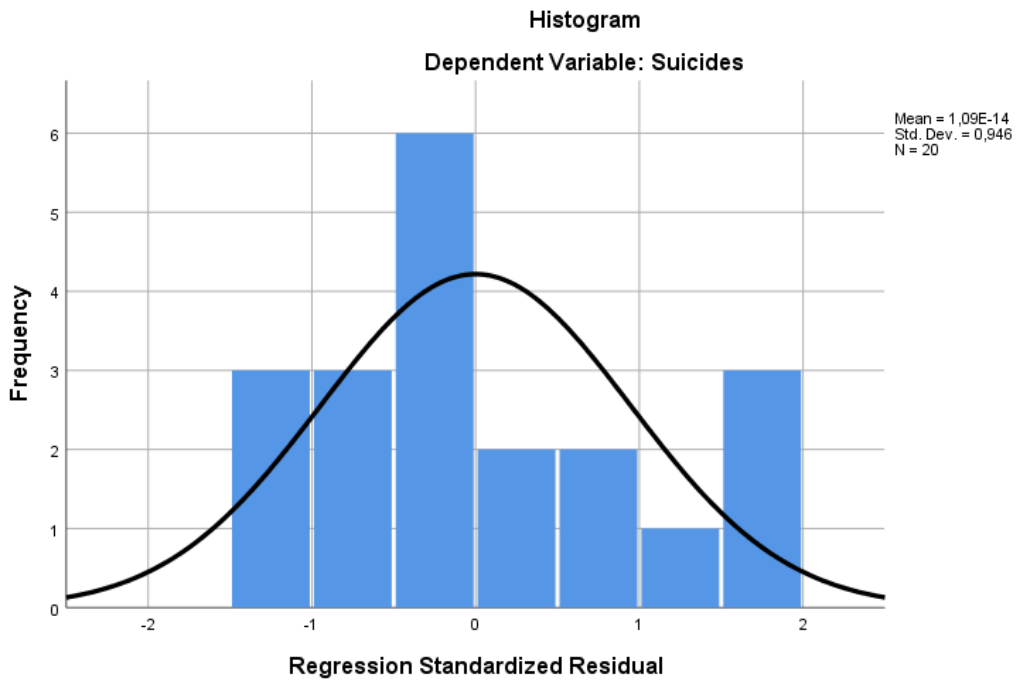
Interpretations

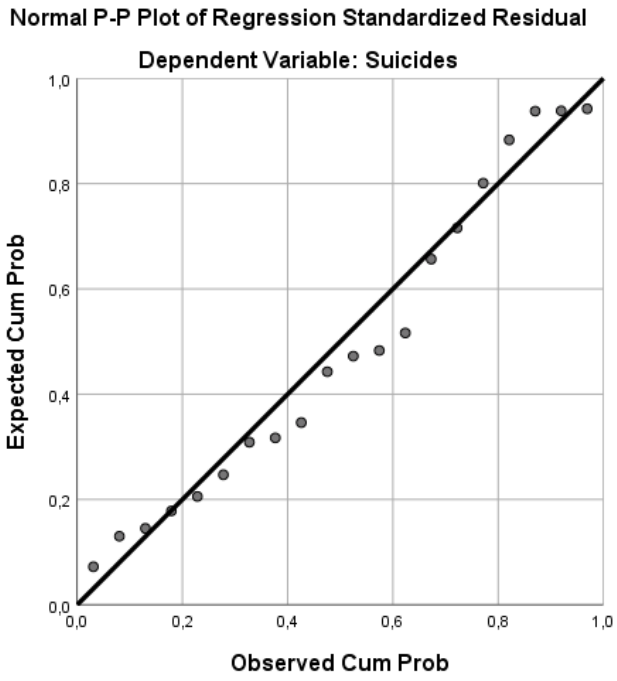
Importance of sampling
It is important to divide the database randomly to be able to test the obtained model. Thus the database in two, the training one and the test one. To assess the quality of the model or predictor

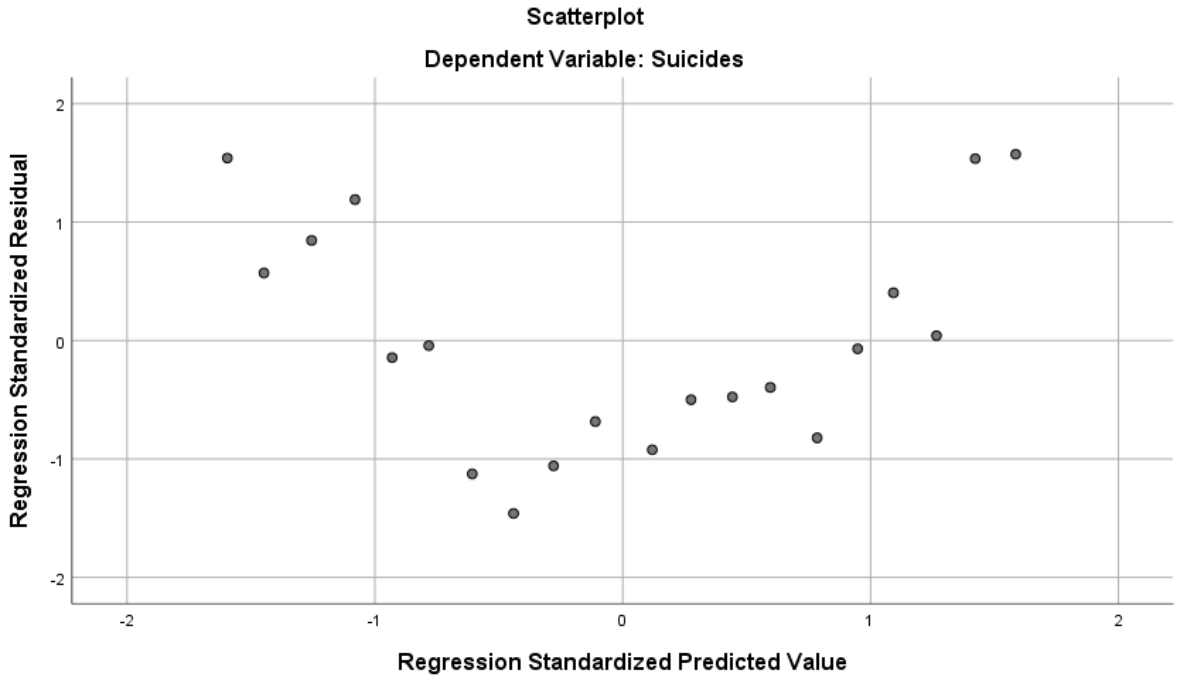
Regression fallacy The Regression Fallacy occurs when one mistakes regression to the mean, which is a statistical phenomenon, for a causal relationship.

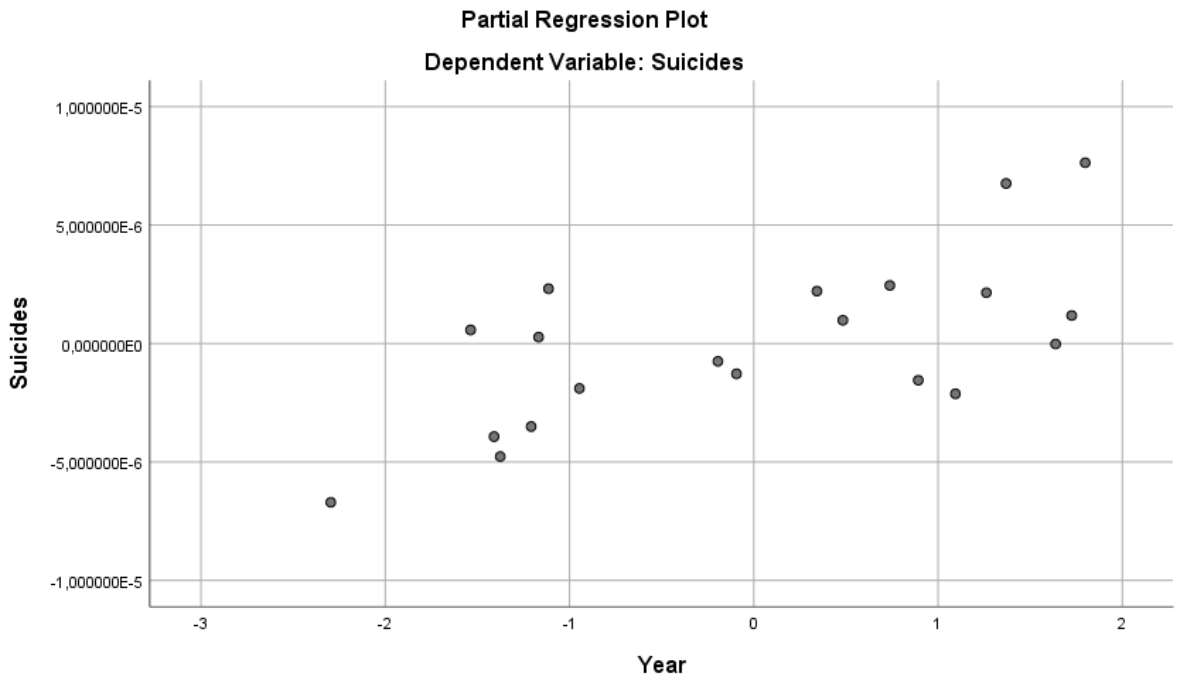
Below is the results of the regression model obtained

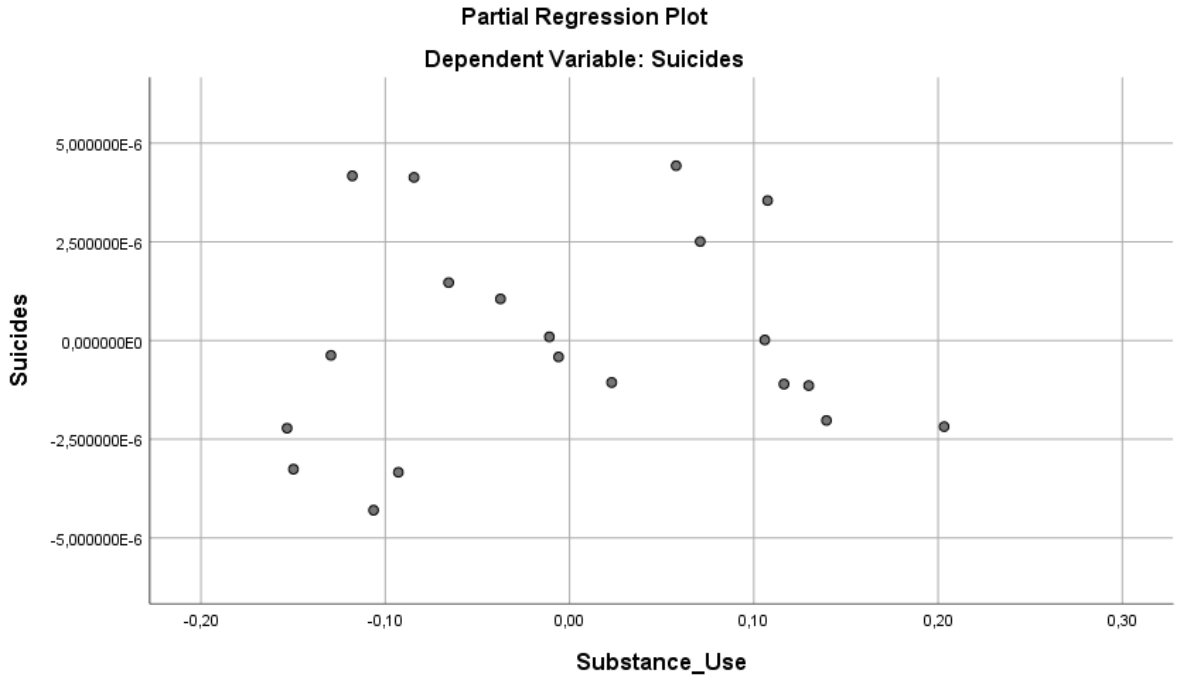
Charts




Normal P-P Plot of Regression Standardized Residual
Dependent Variable: Suicides
Expected Cum Prob
Observed Cum Prob
0,0
0,2
0,4
0,6
0,8
1,0


Scatterplot
Dependent Variable: Suicides
Regression Standardized Residual
Regression Standardized Predicted Value
-2
-1
0
1
2


Partial Regression Plot
Dependent Variable: Suicides
Suicides
1,000000E-5
5,000000E-6
0,000000E0
-5,000000E-6
-1,000000E-5
-3
-2
-1
0
1
2
Year


Partial Regression Plot
Dependent Variable: Suicides
Suicides
5,000000E-6
2,500000E-6
0,000000E0
-2,500000E-6
-5,000000E-6
-0,20
-0,10
0,00
0,10
0,20
0,30
Substance_Use